

UNIVERSITY OF TECHNOLOGY - VNUHCM	HOMEWORK 3	Semester/Academic year	1 / 2026-2027
FACULTY OF COMPUTER SCIENCE AND ENGINEERING	Neural Networks: Representation, Learning Dynamics, and Diagnostics	Due	07/10/2026 07:00
	CO3117 - Machine Learning	Format	Formative take-home

Purpose. Build practical and mathematical intuition for perceptrons/MLPs, activations, gradients, and learning behaviour without mirroring midterm prompts.

- Complete the work by hand where calculation or derivation is requested. Show reasoning; unsupported final numbers receive limited credit.
- Use lecture notes and course resources. Do not use AI, online solution lookup, or classmates' answers for your first attempt. Later supportive AI use must follow the course AI_USE.md protocol.
- These activities are designed for concept transfer and assignment support. They intentionally do not reproduce the wording, ordering, numerical data, or question structure of the midterm.
- No official worked solutions will be released. Students are encouraged to self-check their handwriting with course materials, discussion, and AI assistants after completion.
- Submit one PDF in your course repository and preserve your first attempt. Corrections may be added later without replacing the original submission.

Activity 1 - One perceptron update (1.5 pts)

Use bias input $x_0=1$ and outputs in $\{-1,+1\}$. Current weights are $(0.1,-0.3,0.4)$, learning rate $\eta=0.2$, and one example is $(x_1,x_2)=(1,-1)$ with target -1 . Compute the current prediction and, if needed, one perceptron update.

Activity 2 - Activation functions as design choices (2.0 pts)

- For z in $\{-3,-0.5,0,2\}$, compute ReLU and sigmoid outputs. Then explain how saturation, zero gradients, and output range differ between the two activations.
- Give one situation in which tanh could be preferable to sigmoid in a hidden layer.

Activity 3 - Trace a tiny backpropagation step (2.5 pts)

Consider $x=1$, one hidden sigmoid unit with weight $w=0.6$, one sigmoid output with weight $v=-0.7$, zero biases, target $t=0$, and squared-error loss $0.5(t-o)^2$. Perform the forward pass, compute the output-layer error signal, and determine the sign (increase or decrease) of the gradient update for v and w . Exact final decimals are less important than a correct chain of reasoning.

Activity 4 - Diagnose training curves (1.5 pts)

Sketch or describe three pairs of training/validation-loss curves representing: underfitting, healthy convergence, and overfitting. For each, name one intervention you would try.

Activity 5 - Assignment bridge: controlled capacity experiment (2.5 pts)

- On your longitudinal-assignment dataset, compare two MLPs that differ in only one capacity-related choice (e.g., hidden width, number of hidden layers, or regularization strength). Keep preprocessing, split, metric, and random seed policy fixed.
- Before running, write a prediction about training score, validation score, and runtime. After running, report whether the result supports your prediction and give one plausible explanation for any mismatch.

Submission checklist

- ☐ Name and student ID on first page
- ☐ All calculations show intermediate steps
- ☐ Assignment-bridge activity uses your own dataset/use case
- ☐ First attempt preserved
- ☐ Any later AI/reference use disclosed according to course policy